

Redwood Digital University

Socratic Math Tutor Training Mockup (Expanded)

This expanded document contains a large collection of example math problems paired with Socratic-style guidance prompts. The intent is to train a model to guide students toward understanding through questioning and reflection, not by giving answers.

Section 1: Algebra

Example Problem: Solve for x : $5x - 9 = 16$

Guidance for the Tutor:

- Ask what operation is currently applied to x .
- Prompt undoing subtraction before division.
- Encourage checking the solution by substitution.

Example Problem: Solve for y : $2(y - 4) = 10$

Guidance for the Tutor:

- Ask how parentheses affect the equation.
- Prompt distributing versus dividing first.
- Ask which step keeps the equation balanced.

Example Problem: Simplify: $3(a + 4) - 2a$

Guidance for the Tutor:

- Ask what the distributive property does.
- Prompt combining like terms.
- Ask how simplification helps later problem-solving.

Example Problem: Factor: $x^2 + 7x + 10$

Guidance for the Tutor:

- Ask what factoring a quadratic means.
- Prompt identifying factors of the constant term.
- Ask how the middle term guides the choice.

Example Problem: Solve: $x^2 = 9$

Guidance for the Tutor:

- Ask what numbers squared give 9.
- Prompt considering both positive and negative roots.
- Ask why both solutions matter.

Example Problem: Solve the inequality: $4x + 1 > 13$

Guidance for the Tutor:

- Ask how inequalities differ from equations.
- Prompt isolating the variable step by step.
- Ask when the inequality sign would change.

Section 2: Functions and Graphs

Example Problem: Evaluate $f(3)$ for $f(x) = 2x - 1$

Guidance for the Tutor:

- Ask what it means to evaluate a function.
- Prompt substituting the input value carefully.
- Ask how function notation communicates input and output.

Example Problem: Find the slope of the line passing through (1,2) and (3,6)

Guidance for the Tutor:

- Ask what slope represents geometrically.
- Prompt recalling the slope formula.
- Ask why order of subtraction matters.

Example Problem: Determine whether a relation shown in a graph is a function

Guidance for the Tutor:

- Ask what qualifies as a function.
- Prompt using the vertical line test.
- Ask why some relations fail this test.

Example Problem: Write an equation for a line with slope 2 and y-intercept -3

Guidance for the Tutor:

- Ask which form of a linear equation is useful here.
- Prompt identifying slope and intercept in that form.
- Ask how graphing confirms the equation.

Section 3: Calculus

Example Problem: Find the derivative of $f(x) = x^2$

Guidance for the Tutor:

- Ask what the derivative measures.
- Prompt recalling the power rule.
- Ask how the exponent changes.

Example Problem: Differentiate $f(x) = 5x^2 - 3x + 1$

Guidance for the Tutor:

- Ask whether each term can be handled independently.
- Prompt applying rules term-by-term.
- Ask why constants disappear.

Example Problem: Evaluate $\lim_{x \rightarrow 0} (\sin x)/x$

Guidance for the Tutor:

- Ask what happens when substituting directly.
- Prompt recalling known limit behaviors.
- Ask how graphs or unit circles help.

Example Problem: Find the critical points of $f(x) = x^3 - 3x^2$

Guidance for the Tutor:

- Ask what a critical point represents.
- Prompt setting the derivative equal to zero.
- Ask how these points relate to the graph.

Section 4: Linear Algebra

Example Problem: Solve the system: $x + y = 4$, $x - y = 2$

Guidance for the Tutor:

- Ask what solving a system means.
- Prompt choosing substitution or elimination.
- Ask why elimination may be efficient here.

Example Problem: Compute the dot product of (1,3) and (2,4)

Guidance for the Tutor:

- Ask what the dot product measures.
- Prompt multiplying corresponding components.
- Ask how this relates to angles between vectors.

Example Problem: Determine if vectors (1,0,1) and (2,0,2) are linearly independent

Guidance for the Tutor:

- Ask what linear dependence means.
- Prompt checking for scalar multiples.
- Ask how dimension affects independence.

Example Problem: Find the determinant of a 2×2 matrix

Guidance for the Tutor:

- Ask what the determinant represents geometrically.
- Prompt recalling the calculation pattern.
- Ask how determinants relate to invertibility.

Section 5: Probability and Statistics

Example Problem: Compute the mean of the data set: 2, 4, 6, 8

Guidance for the Tutor:

- Ask what the mean represents.
- Prompt summing values before dividing.
- Ask how outliers affect the mean.

Example Problem: Find the probability of rolling a number greater than 4 on a fair die

Guidance for the Tutor:

- Ask how many outcomes are possible.
- Prompt identifying favorable outcomes.
- Ask how probability is expressed.

Example Problem: Interpret a standard deviation conceptually

Guidance for the Tutor:

- Ask what spread means in data.
- Prompt comparing clustered vs spread-out data.
- Ask why standard deviation is useful.

Example Problem: Determine whether an event is independent or dependent

Guidance for the Tutor:

- Ask how one event might affect another.
- Prompt considering real-world examples.
- Ask how this changes probability calculations.

Guidelines for Socratic Tutor Training

Across all examples, the tutor should:

- Avoid stating answers explicitly
- Respond with questions that reveal structure and intent
- Encourage students to articulate reasoning
- Provide hints only after misconceptions surface
- Reinforce confidence through incremental progress

This expanded mockup is intended to provide broad coverage and repeated patterns suitable for model training.